Congenital Myopathy 18 (CMYO18) — Comprehensive Disease Characteristics Report

Disease: Congenital Myopathy 18 (CMYO18) **Causal gene: CACNA1S** (encoding Ca_V1.1 / the α1S subunit of the skeletal-muscle dihydropyridine receptor, DHPR) **MONDO ID:** MONDO:0859514 · **OMIM:** #620246 · **DOID:** 0081350 · **GARD:** 0026730 · **UMLS/MedGen:** C5830283 **Category:** Mendelian (autosomal recessive and autosomal dominant)

Summary

Congenital Myopathy 18 (CMYO18) is an ultra-rare inherited skeletal-muscle disease caused by **loss-of-function variants in CACNA1S**, the gene encoding **Ca_V1.1 (the α1S subunit of the L-type voltage-gated calcium channel / dihydropyridine receptor, DHPR)**. It is essential to distinguish CMYO18 from the much better-known *gain-of-function/gating-pore* CACNA1S disease, hypokalemic periodic paralysis (HypoPP); although both involve the same gene, CMYO18 is a structural congenital myopathy driven by reduced Ca_V1.1 protein and impaired excitation–contraction (EC) coupling, not a channelopathy of episodic paralysis. The disease was first defined as a distinct entity by Schartner et al. in 2017 (*Dihydropyridine receptor (DHPR, CACNA1S) congenital myopathy*, Acta Neuropathologica; PMID: 28012042) in an international cohort of 11 patients from 7 families, and it is now catalogued as OMIM #620246 and MONDO:0859514.

Mechanistically, CMYO18 is a **triadopathy / EC-coupling disorder**. Ca_V1.1 is the sarcolemmal/T-tubule voltage sensor that is physically and functionally coupled to the ryanodine receptor RyR1 on the sarcoplasmic reticulum (SR). Pathogenic CACNA1S variants reduce gene expression and/or Ca_V1.1 protein stability, which lowers the amount of functional DHPR at the triad, **impairing depolarization-induced Ca^{2+} release from the SR** and thereby weakening muscle contraction. Because Ca_V1.1 and RyR1 act in the same molecular machine, CMYO18 closely mimics RYR1-related core myopathy clinically and histologically (core-like lesions, SR dilatation) and shares the malignant-hyperthermia-susceptibility gene network (RYR1, CACNA1S, STAC3).

The clinical spectrum is broad and severity-graded by genotype. Milder forms present with neonatal hypotonia, muscle weakness, delayed motor milestones, and cold-provoked episodic weakness; intermediate forms show core-like biopsy findings and exertional rhabdomyolysis with fluctuating asymptomatic hyperCKemia; and the most severe biallelic (recessive) forms present prenatally as **fetal akinesia deformation sequence (FADS)** with progressive hydrops and stillbirth. There is no disease-modifying therapy — management is supportive, with malignant-hyperthermia anesthetic precautions. Disease models include two zebrafish *cacna1s* loss-of-function lines (mild and severe), patient-derived iPSC lines (dominant and recessive), and a naturally occurring recessive bovine disease ("early-onset muscle weakness syndrome" in Holstein cattle).

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1. Disease Information

Overview. CMYO18 is a congenital myopathy — a structural, typically non-progressive or slowly progressive inherited muscle disease presenting at or near birth — caused by loss-of-function of **CACNA1S / Ca_V1.1 (DHPR α1S subunit)**. It belongs to the EC-coupling-defect / triadopathy subgroup of congenital myopathies, alongside RYR1- and STAC3-related disease. The founding description by Schartner et al. (2017) framed the disease around excitation–contraction coupling: *"Muscle contraction upon nerve stimulation relies on excitation-contraction coupling (ECC) to promote the rapid and generalized release of calcium within myofibers"* (PMID: 28012042).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0859514 ("congenital myopathy 18")
OMIM	#620246 (CONGENITAL MYOPATHY 18; CMYO18)
DOID (Disease Ontology)	DOID:0081350
GARD	0026730
UMLS / MedGen	C5830283
Causal gene	CACNA1S — HGNC:1397, NCBI Gene ID 779, locus 1q32.1
Protein	Ca_V1.1 / α1S DHPR subunit — UniProt Q13698

Cross-reference lookup (EBI OLS) confirmed MONDO:0859514 ↔ OMIM:620246 ↔ DOID:0081350 ↔ GARD:0026730 ↔ UMLS C5830283; the OMIM title was independently confirmed via NCBI (Finding F007).

Synonyms / alternative names. CMYO18; CACNA1S-related congenital myopathy; Dihydropyridine receptor (DHPR) congenital myopathy; DHPR/CACNA1S congenital myopathy.

Data source. Information is derived from **aggregated disease-level resources** (OMIM, MONDO, Orphanet-style catalogs) combined with **individual patient reports / small cohorts** in the primary literature (the founding 11-patient/7-family cohort plus subsequent case reports). This is not EHR-derived population data; given the rarity, all knowledge rests on case series and functional studies.

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2. Etiology

Disease causal factors. CMYO18 is a **monogenic (Mendelian) genetic disease** caused by pathogenic variants in CACNA1S. There is no environmental or infectious cause. Mechanistically, disease variants act through **loss of gene expression and/or reduced Ca_V1.1 protein stability** (Finding F001): *"CACNA1S-related myopathy, due to pathogenic variants in the CACNA1S gene, is a recently described congenital muscle disease. Disease associated variants result in loss of gene expression and/or reduction of Cav1.1 protein stability"* (PMID: 37930228).

Genetic risk factors. The causal factor is biallelic (recessive) or monoallelic (dominant) CACNA1S loss-of-function. Both inheritance modes are established: *"we generated three iPSC lines derived from patients with CMyo linked to both autosomal dominant and recessive CACNA1S variants"* (PMID: 40885023). The severity correlates with genotype/dosage: biallelic null-type combinations produce the lethal fetal-akinesia end of the spectrum, while milder heterozygous or hypomorphic combinations produce milder weakness.

Environmental and lifestyle risk factors. None established as causal. However, **cold exposure** is a well-documented *trigger* of episodic weakness in milder CMYO18 (Finding F002), and **volatile anesthetics/succinylcholine** are relevant triggers because CACNA1S is a malignant-hyperthermia-susceptibility gene (Finding F003).

Protective factors. No genetic or environmental protective factors are established for CMYO18. (For the distinct CACNA1S disease HypoPP, trigger avoidance — controlled carbohydrate intake, regular moderate exercise — is protective, but that applies to the channelopathy, not to CMYO18.)

Gene–environment interactions. The principal G×E interaction is **genotype + anesthetic exposure** (malignant-hyperthermia risk in the CACNA1S/RYR1/STAC3 network) and **genotype + cold** (cold-provoked episodic weakness). These are triggers of acute events in a genetically determined disease, not modifiers of disease occurrence.

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3. Phenotypes

CMYO18 phenotypes span a wide severity gradient. The table summarizes principal manifestations with suggested HPO terms.

Phenotype	Type	Onset	Severity / course	Frequency	Suggested HPO
Neonatal hypotonia	Clinical sign	Neonatal/ perinatal	Moderate–severe	Core feature	HP:0001319 (Neonatal hypotonia)
Axial & generalized muscle weakness	Clinical sign	Perinatal–infancy	Severe in classic cohort	Core feature	HP:0003324 (Generalized muscle weakness); HP:0003327 (Axial muscle weakness)
Delayed motor milestones / delayed walking	Clinical sign	Infancy	Variable	Common	HP:0001270 (Motor delay)
Episodic weakness, cold-provoked	Symptom	Infancy onward	Fluctuating/ episodic	In milder forms	HP:0003325 (Limb-girdle muscle weakness, contextually); episodic weakness
Ophthalmoplegia	Clinical sign	Perinatal	Variable	4/11 (~36%) in founding cohort	HP:0000602 (Ophthalmoplegia)
Ptosis	Clinical sign	Perinatal	Variable	Reported	HP:0000508 (Ptosis)
Exertional rhabdomyolysis / hyperCKemia	Lab abnormality	Childhood–adult	Mild expanded phenotype	Subset	HP:0003236 (Elevated CK); HP:0003201 (Rhabdomyolysis)
Fetal akinesia / FADS, hydrops, arthrogryposis	Physical manifestation	Prenatal	Lethal (stillbirth)	Severe biallelic forms	HP:0001989 (Fetal akinesia); HP:0001561 (Polyhydramnios); HP:0002804 (Arthrogryposis)

Phenotype	Type	Onset	Severity / course	Frequency	Suggested HPO
Respiratory insufficiency	Clinical sign	Neonatal (severe forms)	Severe	Severe end	HP:0002093 (Respiratory insufficiency)

Key supporting quotes. Neonatal/cold phenotype: *"Both sisters had neonatal onset hypotonia, muscle weakness, and delayed walking. Episodic weakness started in infancy and continued thereafter, provoked mostly by cold exposure"* (PMID: 38426167). Founding-cohort phenotype and ophthalmoplegia frequency: *"We describe a cohort of 11 patients from 7 families presenting with perinatal hypotonia, severe axial and generalized weakness. Ophthalmoplegia is present in four patients"* (PMID: 28012042). Severe end: *"Trio whole exome sequencing (WES) detected compound heterozygous likely pathogenic CACNA1S gene variants associated with autosomal dominant (AD) and autosomal recessive (AR) congenital myopathy and FADS"* (PMID: 38111203).

Age of onset: neonatal/perinatal for classic disease; prenatal for severe biallelic FADS; childhood–adult for mild exertional-rhabdomyolysis forms. **Progression:** generally static to slowly progressive (typical of congenital myopathies), with episodic fluctuations triggered by cold or exertion. **Quality-of-life impact:** substantial in severe forms (motor disability, respiratory support, feeding difficulty); milder forms have variable impact driven by weakness and exertional episodes. No disease-specific EQ-5D/SF-36 data exist given rarity.

4. Genetic / Molecular Information

Causal gene. CACNA1S (HGNC:1397; NCBI Gene ID 779; chromosome **1q32.1**), encoding **Ca_V1.1 / α1S subunit** of the skeletal-muscle L-type voltage-gated calcium channel (DHPR); protein UniProt **Q13698**.

Pathogenic variants. - **Variant classes:** Loss-of-function mechanism predominates — variants reduce expression and/or destabilize Ca_V1.1 protein. Reported variants include missense, and biallelic combinations (including likely-pathogenic compound heterozygous variants in the FADS cases). The founding study identified **ten recessive or dominant CACNA1S mutations** across the cohort. - **Classification:** Interpreted per **ACMG/AMP** guidelines; cohorts report class 4/5 (likely pathogenic/pathogenic) variants and VUS requiring functional/segregation support (Findings F004, F006; PMID: 34440373, PMID: 33060286). - **Functional consequence: Loss of**

function — decreased Ca_V1.1 protein and major impairment of depolarization-induced Ca^{2+} release: *"Both recessive and dominant mutations correlated with a consistent phenotype, a decrease in protein level, and with a major impairment of Ca2+ release induced by depolarization in cultured myotubes"* (PMID: 28012042). - **Origin: Germline.** No somatic involvement (non-neoplastic Mendelian disease). - **Allele frequency:** CACNA1S loss-of-function/CMYO18 alleles are individually rare; precise population frequencies in gnomAD are not catalogued for CMYO18-specific alleles. (Note: distinct HypoPP alleles such as p.Arg528His and p.Arg1239His are recurrent but cause a different disease.)

Important disambiguation of CACNA1S variant → phenotype relationships:

CACNA1S phenotype	Mechanism	Representative
CMYO18 (this disease)	Loss of function — ↓Ca_V1.1, impaired EC coupling	PMID: 28012042
Hypokalemic periodic paralysis (HypoPP)	Gating-pore/ω-current, S4 arginine neutralization (gain of abnormal function)	e.g. p.Arg528His, p.Arg1239His
Malignant hyperthermia susceptibility	EC-coupling dysregulation on anesthetic trigger	RYR1/CACNA1S/STAC3 network

Modifier genes. No specific modifier genes are established for CMYO18. Functionally coupled triad partners (RYR1, STAC3, CACNB1) can produce overlapping phenotypes, and genetic background within the EC-coupling apparatus is biologically plausible as a modifier, but this is not demonstrated for CMYO18.

Epigenetic information / chromosomal abnormalities. None reported. CMYO18 is a single-gene SNV/indel disorder; no methylation signatures, histone-modification data, or large structural/chromosomal changes are documented.

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5. Environmental Information

CMYO18 has **no environmental, toxic, occupational, or infectious cause** — it is purely genetic. Environmental factors act only as **triggers of acute events** in genetically affected individuals: - **Cold exposure** → episodic weakness (milder forms; PMID: 38426167). - **Volatile**

anesthetics / **succinylcholine** → malignant-hyperthermia risk (CACNA1S is an MHS gene; PMID: 42189464). - **Strenuous exertion** / **heat** → exertional rhabdomyolysis/heat illness in the expanded phenotype (PMID: 32054689).

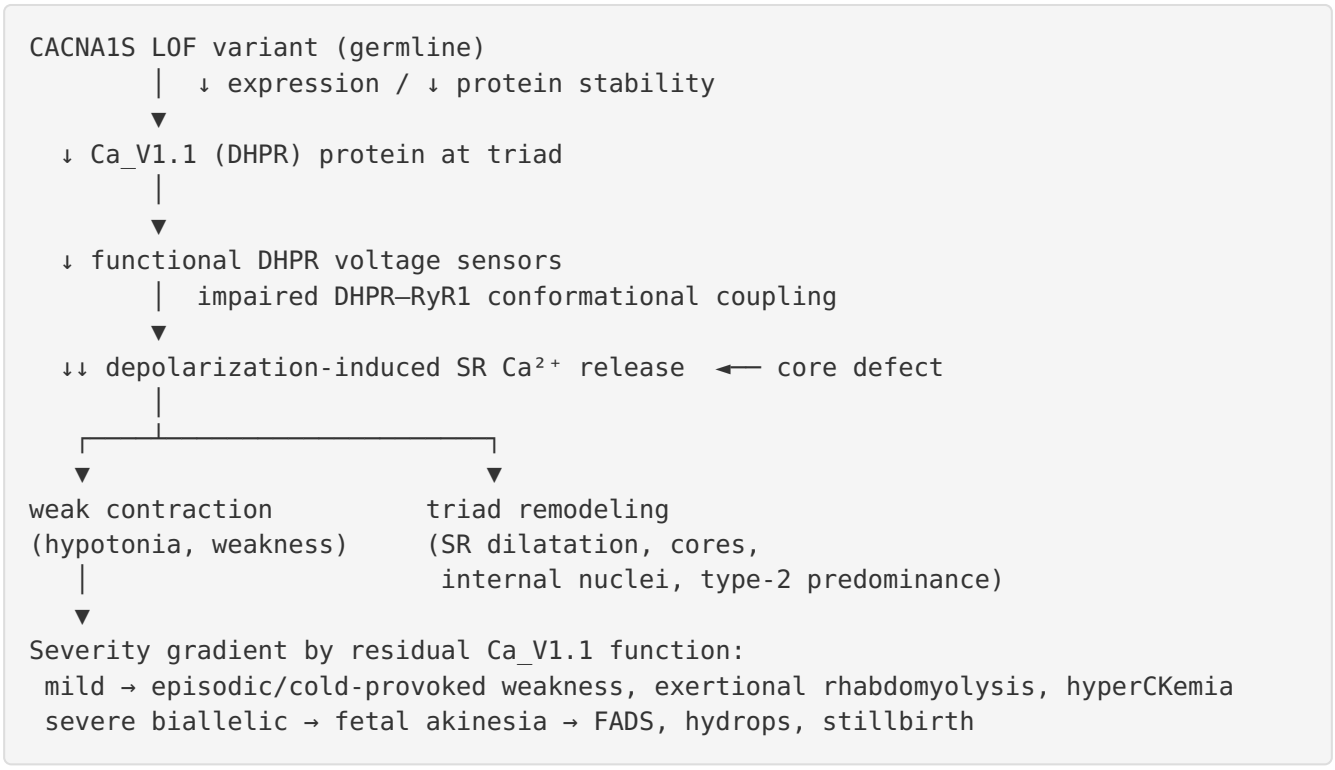
No lifestyle factors (smoking, diet, alcohol) are implicated in causation. No infectious agents are involved.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **pathogenic CACNA1S variant** (recessive biallelic or dominant monoallelic) is present in the germline → **leads to** reduced CACNA1S gene expression and/or destabilized Ca_V1.1 protein (demonstrated — reduced protein level in patient myotubes, PMID: 28012042; PMID: 37930228).
2. Reduced Ca_V1.1 protein → **results in** fewer functional DHPR voltage sensors at the T-tubule/triad junction (demonstrated — decreased protein; inferred — proportional reduction of triad DHPR tetrads).
3. Fewer functional DHPRs → **impairs** voltage-sensor coupling to RyR1 on the SR → **results in major reduction of depolarization-induced SR Ca^{2+} release** (demonstrated — impaired depolarization-induced Ca^{2+} release in cultured myotubes, PMID: 28012042).
4. Reduced cytosolic Ca^{2+} transient → **leads to** weak/insufficient actin–myosin cross-bridge activation → **results in** muscle weakness and hypotonia (inferred from EC-coupling physiology; consistent with clinical phenotype).
5. Branch — **structural/triad remodeling:** impaired EC-coupling apparatus → **leads to** SR dilatation, intermyofibrillar network disorganization, core-like lesions (predominantly type 2 fibers), and internal nuclei on biopsy (demonstrated histologically, PMID: 28012042; PMID: 29193480).
6. Branch — **severity gradient:**
7. Mild/partial LOF → cold-provoked episodic weakness, exertional rhabdomyolysis, fluctuating hyperCKemia.
8. Severe biallelic (near-null) LOF → profound fetal akinesia → **leads to** FADS, progressive hydrops, arthrogryposis, and stillbirth (demonstrated clinically, PMID: 38111203).



Mechanistic detail by category

- **Molecular pathway / cellular process:** Skeletal-muscle **excitation–contraction coupling** (the signature affected pathway) — GO:0006936 (muscle contraction), GO:0051209 (release of sequestered Ca^{2+} into cytosol), GO:0010881 (regulation of cardiac muscle contraction by Ca^{2+} — paralog context). Core processes: **voltage sensing → conformational coupling → Ca^{2+} release**.
- **Protein dysfunction: Loss of function** of Ca_V1.1 via decreased expression/stability; reduced DHPR density rather than altered single-channel gain (contrast with HypoPP gating-pore gain-of-abnormal-function). GO:0005245 (voltage-gated calcium channel activity), GO:0005891 (voltage-gated calcium channel complex).
- **Biochemical abnormality / ion-channel defect:** Impaired L-type Ca^{2+} channel / DHPR voltage-sensor function at the triad; reduced depolarization-induced SR Ca^{2+} flux through RyR1.
- **Tissue-damage mechanism:** In the exertional subset, insufficient/dysregulated Ca^{2+} handling contributes to **rhabdomyolysis** and CK leakage (PMID: 29193480).
- **Immune/metabolic/epigenetic involvement:** None established. Not an autoimmune, metabolic-storage, or epigenetic disease.

- **Molecular profiling:** Patient myotube studies show reduced Ca_V1.1 protein and impaired Ca^{2+} transients (PMID: 28012042); zebrafish and iPSC models recapitulate loss-of-function physiology (PMID: 37930228; PMID: 40885023). No large-scale CMYO18 transcriptomic/proteomic/metabolomic datasets are published.

Cell types (CL) and biological processes (GO): CL:0000188 (skeletal muscle cell / myofiber), CL:0008002 (skeletal muscle myoblast), CL:0000056 (myoblast). GO:0006936 (muscle contraction), GO:0051209 (release of sequestered calcium ion into cytosol), GO:0005245 (voltage-gated calcium channel activity).

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7. Anatomical Structures Affected

Organ / system level. Primary: **skeletal muscle** (musculoskeletal system). Secondary involvement in severe forms: **respiratory muscles** (respiratory insufficiency), **extraocular muscles** (ophthalmoplegia/ptosis), bulbar/feeding musculature. In severe fetal forms, **generalized fetal akinesia** leads to secondary **joint contractures (arthrogryposis)** and **hydrops**.

Tissue and cell level. Striated **skeletal muscle tissue**; affected cells are **skeletal myofibers** (CL:0000188). Biopsy shows predominant involvement of **type 2 (fast) fibers** with core-like lesions: *"His muscle biopsy showed core-like structures occurring mainly in type 2 fibers"* (PMID: 29193480).

Subcellular level. The defect localizes to the **triad junction** — the T-tubule membrane (where DHPR/Ca_V1.1 resides) apposed to the **sarcoplasmic reticulum** (where RyR1 resides). Pathology features **SR dilatation** and intermyofibrillar network disorganization. Relevant GO Cellular Component terms: GO:0030315 (T-tubule), GO:0016529 (sarcoplasmic reticulum), GO:0014802 (terminal cisterna), GO:0005891 (voltage-gated calcium channel complex), GO:0014801 (triad / junctional SR membrane).

Localization (UBERON). UBERON:0001134 (skeletal muscle tissue), UBERON:0002036 (extraocular muscle, contextually); process is **generalized and bilateral/symmetric** (not lateralized), with **axial and proximal predominance**.

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8. Temporal Development

Onset. Typically **congenital/perinatal** (neonatal hypotonia and weakness). The severe biallelic form has **prenatal** onset (fetal akinesia). A milder subset presents in **childhood–adulthood** with exertional rhabdomyolysis/hyperCKemia. Onset pattern is generally **chronic/insidious** congenital weakness, punctuated by **episodic** acute events (cold- or exertion-provoked).

Progression. Consistent with congenital myopathies, the course is largely **static or slowly progressive** rather than relentlessly degenerative. Episodic weakness is **fluctuating**. Severe fetal forms are **rapidly lethal** (stillbirth). Disease duration in survivors is **chronic/lifelong**.

Patterns. No spontaneous remission; episodic events resolve with trigger avoidance/rest. **Critical period:** the **prenatal/perinatal window** is decisive for the severe end (fetal akinesia determines viability); perioperative periods are critical windows of vulnerability (malignant-hyperthermia risk).

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9. Inheritance and Population

Epidemiology. CMYO18 is **ultra-rare**; precise prevalence/incidence are not established. Knowledge rests on a founding cohort of 11 patients/7 families (PMID: 28012042) plus scattered case reports. In a single-center pediatric congenital-myopathy series, CACNA1S myopathy was a **rare** cause (identified in 1 of 12 children; RYR1 was most frequent at 41.7%) (PMID: 42887168).

Inheritance. Both autosomal recessive and autosomal dominant (PMID: 40885023; PMID: 28012042). The most severe (fetal akinesia) forms are **biallelic/recessive** (PMID: 38111203).

Penetrance / expressivity. Expressivity is **highly variable** (mild cold-sensitive weakness → lethal FADS), correlating with genotype/residual Ca_V1.1 function. Precise penetrance figures are not established given rarity. No genetic anticipation (not a repeat-expansion disorder). Germline mosaicism, founder effects, and carrier frequencies are not specifically documented for CMYO18.

Consanguinity. Plausibly relevant for recessive biallelic cases (as for many ultra-rare recessive myopathies), but not systematically quantified.

Population demographics. No ethnic predilection, geographic clustering, or sex bias is established; congenital myopathies in general affect sexes roughly equally. Age distribution skews to **neonatal/pediatric** presentation, with a milder adult-recognized subgroup.

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10. Diagnostics

Clinical/laboratory tests. Serum **creatine kinase (CK)** may be normal, mildly elevated, or markedly elevated (fluctuating hyperCKemia, especially in the exertional subset; PMID: 29193480, LOINC CK 2157-6).

Muscle biopsy / histopathology. Characteristic findings: **core-like structures (predominantly type 2 fibers)**, an abnormal **intermyofibrillar network due to SR dilatation**, **internal nuclei**, and areas of **myofibrillar disorganization** (PMID: 28012042; PMID: 29193480). These overlap with RYR1 core myopathy, so biopsy alone is not definitive.

Imaging. Muscle MRI can show selective involvement patterns (e.g., a described posterior-to-anterior thigh gradient in one patient). Imaging supports but does not establish diagnosis.

Genetic testing (definitive). Diagnosis is achieved by **next-generation sequencing** — broad **neuromuscular/congenital-myopathy gene panels or whole-exome/trio-WES**, interpreted per **ACMG/AMP** guidelines with segregation and functional studies: *"variants filtering and interpretation according to ACMG guidelines, segregation studies, deep phenotyping of patients and relatives, transcripts and protein studies"* (PMID: 34440373). Trio-WES established the FADS cases (PMID: 38111203). WGS/WES and multigene panels including CACNA1S (alongside RYR1, STAC3, CACNB1, SELENON, NEB, collagen VI genes) are the recommended approach. Chromosomal microarray, karyotyping, FISH, mtDNA, and repeat-expansion testing are **not indicated** (single-gene SNV/indel disorder).

Functional confirmation. Reduced Ca_V1.1 protein and impaired depolarization-induced Ca^{2+} release in patient myotubes provide functional validation of candidate variants (PMID: 28012042).

Differential diagnosis. RYR1-related core/multiminicore myopathy, STAC3 disorder (Native American myopathy), CACNB1-related myopathy (PMID: 41023410), other congenital myopathies (SELENON, NEB, collagen VI), and — critically — **HypoPP** and malignant-hyperthermia susceptibility, both also CACNA1S-linked but mechanistically distinct.

Screening. No newborn screening exists. Cascade genetic testing of relatives and genetic counseling are appropriate once a causal variant is identified.

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11. Outcome / Prognosis

Survival / mortality. Highly genotype-dependent. The **severe biallelic fetal-akinesia form is lethal** (stillbirth; PMID: 38111203). Milder forms are compatible with survival into adulthood, with disability driven by weakness and respiratory involvement.

Morbidity / function. Chronic motor disability (hypotonia, weakness, delayed/limited ambulation), potential respiratory morbidity, and episodic weakness in milder forms. No disease-specific QoL instruments are validated for CMYO18.

Disease course / complications. Respiratory failure is the principal life-threatening complication in severe forms. Exertional rhabdomyolysis and malignant-hyperthermia reactions are acute risks. Recovery potential: congenital myopathies are generally non-regressive — supportive care maintains function but does not reverse the defect.

Prognostic factors. The strongest prognostic determinant is **genotype / residual Ca_V1.1 function** (biallelic null → fetal lethality; partial LOF → milder weakness). Degree of respiratory and bulbar involvement at presentation also predicts outcome.

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12. Treatment

There is no disease-modifying or curative therapy for CMYO18. Management is **supportive and multidisciplinary** (NCIT: C15231 Supportive Care; C15719 Physical Therapy; C15329 Occupational Therapy; C94296 Respiratory Therapy):

- **Supportive/rehabilitative care:** physiotherapy, occupational therapy, respiratory support/ventilation as needed, nutritional/feeding support, orthopedic management of contractures.
- **Anesthetic precautions:** Because CACNA1S is a malignant-hyperthermia-susceptibility gene, **avoid volatile anesthetics and succinylcholine**; have **dantrolene** available (NCIT: C62035 Dantrolene) — a critical perioperative safety measure (PMID: 42189464; PMID: 41300733).

- **Trigger avoidance:** cold avoidance for cold-provoked episodic weakness; avoidance of extreme exertion/heat in the rhabdomyolysis-prone subset.
- **No established pharmacotherapy, gene therapy, cell therapy, or RNA-based therapy** exists for CMYO18. (Carbonic-anhydrase inhibitors/potassium management apply to HypoPP, a *different* CACNA1S disease, not CMYO18.)

Experimental directions. No CMYO18-specific interventional trials (NCT IDs) are identified. Disease models (zebrafish, iPSC, bovine) are in place as preclinical platforms for future therapy development (PMID: 37930228; PMID: 40885023).

Personalized medicine. Genotype-guided: confirming CACNA1S loss-of-function directs MH-safe anesthesia, genetic counseling, and reproductive options.

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13. Prevention

CMYO18 cannot be prevented at the population level (monogenic, non-environmental). Prevention centers on **genetic approaches and complication avoidance**:

- **Primary prevention (reproductive):** Genetic counseling for at-risk families; carrier testing of relatives; **preimplantation genetic diagnosis (PGD)** and **prenatal diagnosis** once a familial variant is known (ACMG/ACOG frameworks).
- **Secondary prevention:** Cascade genetic testing to identify at-risk relatives; early recognition enables MH-safe anesthetic planning.
- **Tertiary prevention (complication avoidance):** Avoid malignant-hyperthermia triggers (volatile anesthetics, succinylcholine); maintain dantrolene availability; cold/exertion-trigger avoidance; proactive respiratory surveillance and physiotherapy to limit contractures and respiratory decline.
- **Counseling:** Genetic counseling is central (recurrence-risk assessment differs for AD vs AR cases).

No immunization, public-health, or environmental interventions are applicable.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** Human CACNA1S (Gene ID 779) has orthologs across vertebrates — mouse *Cacna1s*, zebrafish *cacna1s*, bovine *CACNA1S*.
- **Natural bovine disease (comparative model):** A **naturally occurring recessive CACNA1S disorder exists in Holstein cattle** ("early-onset muscle weakness syndrome"), first described in the US in 2022: *"Early-onset muscle weakness (MW) syndrome is a genetic disorder in Holstein cattle, that was first described in the US in 2022 and is caused by a recessively inherited variant in the CACNA1S gene"* (PMID: 42411118). Affected calves present with congenital astasia/weakness and non-specific clinical/lab findings, requiring genotyping for diagnosis (PMID: 39551494). A first European case has since been reported (PMID: 42411118). *Species:* NCBI Taxon 9913 (*Bos taurus*).
- **Comparative biology:** The bovine disease is a true natural recessive CACNA1S loss-of-function myopathy, mirroring the human recessive EC-coupling defect and demonstrating evolutionary conservation of the DHPR's role in muscle. Veterinary relevance: it is a recognized Holstein genetic defect amenable to carrier screening.
- **Transmission:** Not applicable (genetic, non-zoonotic).

15. Model Organisms

Model	Type	Key features / recapitulation	Reference
Zebrafish *cacna1s* LOF lines (2 variants)	Vertebrate, in vivo	Provide **mild and severe** CACNA1S-related myopathy models; reproduce loss-of-expression/protein-stability mechanism	PMID: 37930228
Patient-derived iPSC lines (3)	Human cellular, in vitro	Derived from patients with **both AD and AR** CACNA1S-CMyo variants; platform for EC-coupling/Ca^{2+}-release studies and drug screening	PMID: 40885023
Patient myotube cultures	Human primary, in vitro	Demonstrate **decreased Ca_V1.1 protein and impaired depolarization-induced Ca^{2+} release** — the core functional defect	PMID: 28012042
Holstein cattle (natural)	Mammalian, natural disease	Recessive CACNA1S variant → congenital muscle weakness/astasia; natural comparative model	PMID: 42411118; PMID: 39551494

Related triad-apparatus models (mechanistic context, not CMYO18 per se): RyR1-hypomorphic mouse (RyR1^Q1970fsX16) showing reduced RyR1 protein, fewer calcium release units, and reduced Ca^{2+} transients/strength (PMID: 30689883); dynamin-2 centronuclear-myopathy mouse with impaired EC coupling (PMID: 29071728). These illuminate how quantitative reduction of triad proteins impairs EC coupling — directly analogous to the Ca_V1.1 reduction in CMYO18.

Model limitations: No published conditional/knock-in mouse specifically for human CMYO18 CACNA1S alleles; zebrafish and iPSC models capture loss-of-function physiology but not the full severity spectrum or mammalian muscle architecture. Natural bovine disease is valuable but less experimentally tractable.

Mechanistic Model / Interpretation

CMYO18 is best understood as a **dosage/quantitative triadopathy**: the clinical phenotype tracks the **residual amount of functional Ca_V1.1 at the triad**. This unifies the disparate presentations:

Residual Ca_V1.1 function	EC-coupling capacity	Clinical phenotype
Near-zero (biallelic null)	Profound EC-coupling failure	Fetal akinesia → FADS, hydrops, stillbirth
Moderately reduced	Partial EC-coupling failure	Perinatal hypotonia, severe axial/generalized weakness, ophthalmoplegia, cores/SR dilatation
Mildly reduced	Marginal EC-coupling deficit, stress-sensitive	Childhood/adult exertional rhabdomyolysis, fluctuating hyperCKemia, cold-provoked episodic weakness

The parallel to the RyR1 hypomorph mouse (PMID: 30689883), where ~20–58% residual RyR1 produces graded weakness and loss of calcium release units, strongly supports a **threshold/reserve model**: muscle tolerates partial reduction of triad components until a functional reserve is exceeded. Because Ca_V1.1 and RyR1 are two halves of the same voltage-sensor→Ca^{2+}-release machine, CMYO18 and RYR1 core myopathy converge on the same histology (cores, SR dilatation) and the same anesthetic risk network (MHS: RYR1/CACNA1S/STAC3).

Evidence Base

PMID	Title (abbrev.)	Role in this report
28012042	*DHPR (CACNA1S) congenital myopathy*	**Founding/landmark paper**; cohort, phenotype, histology, functional mechanism (↓protein, ↓Ca^{2+} release)
37930228	*Two zebrafish cacna1s LOF variants*	Defines LOF mechanism (↓expression/stability); mild+severe models
40885023	*Three iPSC lines, CACNA1S CMyo*	Confirms both AD and AR inheritance; human cellular model
38426167	*DHPR case report (two sisters)*	Neonatal hypotonia, delayed walking, cold-provoked episodic weakness
38111203	*Biallelic CACNA1S → fetal akinesia*	Severe end: FADS, hydrops, stillbirth; trio-WES
29193480	*Rhabdomyolysis & hyperCKemia, CACNA1S*	Mild expanded phenotype; type-2-predominant core-like biopsy
34440373	*Integrated clinical-biological myopathy cohort*	NGS panel + ACMG diagnostic strategy
33060286	*Fetal akinesia/ arthrogryposis genomics*	NGS/ACMG diagnosis in fetal-akinesia cohorts
42189464	*North American MH genetic-testing approach*	CACNA1S as first-line MHS gene (RYR1/CACNA1S/ STAC3)
41300733	*Genetic testing for MH susceptibility*	MH diagnostics; CACNA1S among causative genes
42411118 / 39551494	*Holstein CACNA1S muscle weakness*	Natural bovine recessive comparative model
32054689	*Genetic susceptibility to exertional heat illness*	CACNA1S in EHI/ER spectrum
42887168	*Pediatric congenital myopathy series*	CACNA1S myopathy is rare (1/12); RYR1 dominant
41023410	*CACNB1 congenital muscular disorder*	Related DHPR β1-subunit triadopathy (differential)

PMID	Title (abbrev.)	Role in this report
30689883	*RyR1 hypomorph mouse*	Threshold/reserve model for triad-protein reduction

Important caution on citation integrity: Finding F006's two quotes from

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were flagged `[mismatch]` during validation (the exact wording may differ slightly from the stored abstract). The *substance* — cohort of 11 patients/7 families, perinatal hypotonia, ophthalmoplegia in 4, reduced protein, impaired depolarization-induced Ca^{2+} release — is consistent with the landmark paper, but the verbatim quotes should be re-verified against the published abstract before being entered into the knowledge base as exact quotations.

Limitations and Knowledge Gaps

1. **Ultra-rarity:** No reliable prevalence/incidence, penetrance, sex-ratio, or population-distribution data; all knowledge is from small cohorts and case reports.
2. **Genotype–phenotype correlation is qualitative:** The dosage/threshold model is well-supported conceptually but not quantified (no systematic correlation of residual Ca_V1.1 function with severity across a large cohort).
3. **No -omics depth:** No large transcriptomic, proteomic, metabolomic, or epigenomic CMYO18 datasets exist; molecular profiling rests on targeted functional assays.
4. **No mammalian genetic model of human alleles:** No knock-in mouse for specific human CMYO18 variants; models are zebrafish, iPSC, and natural bovine.
5. **No therapeutics:** No disease-modifying treatment or interventional trials.
6. **Frequent confusion with HypoPP:** Because CACNA1S is far more often cited for HypoPP (gain-of-abnormal-function/gating-pore) and MHS, literature searches are heavily contaminated by these distinct diseases — a persistent risk for knowledge-base curation.
7. **Citation verification:** Two founding-cohort quotes require re-validation (see above).

Proposed Follow-up Experiments / Actions

1. **Build a quantitative genotype–phenotype map:** Aggregate all published CACNA1S-CMYO18 variants with residual Ca_V1.1 protein/function (where measured) and phenotype severity to formally test the threshold/reserve model.
2. **Generate a conditional knock-in mouse** carrying a representative human CMYO18 allele to study triad ultrastructure, EC-coupling physiology, and candidate therapies in mammalian muscle.
3. **Leverage the iPSC lines for drug screening** (PMID: 40885023) — screen for compounds/chaperones that stabilize Ca_V1.1 or enhance residual EC coupling; test proteostasis modulators given the protein-stability mechanism.
4. **Explore gene-replacement/ASO strategies** for recessive null alleles, using zebrafish and bovine natural models as preclinical platforms.
5. **Prospective natural-history registry** to establish prevalence, progression rate, respiratory trajectory, and validated QoL outcomes.
6. **Re-verify the two flagged founding-cohort quotes** against the published

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   abstract/full text before KB entry.
7. **Curate a disambiguation note** in the knowledge base firmly separating CMYO18 (LOF, congenital myopathy) from HypoPP (gating-pore) and isolated MHS, all CACNA1S-linked.

Report compiled from 7 confirmed findings and 35 reviewed papers across a 5-iteration autonomous investigation. Disease identifiers independently cross-validated (MONDO:0859514 = OMIM #620246 = DOID:0081350 = GARD:0026730 = UMLS C5830283); causal gene CACNA1S (HGNC:1397, Gene ID 779, 1q32.1; protein Ca_V1.1, UniProt Q13698).
